A MORPHISM FROM AN ALGEBRAIC STACK TO A STACK WITH REPRESENTABLE DIAGONAL IS ALGEBRAIC

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Base change of properties. algebraic-lemma-base-change-representable-by-spaces, algebraic-lemma-base-change-representable-transformations-property (cross-chapter, named). Transfers representability, surjectivity and smoothness of U -> X to the base-changed morphism.
- Representable diagonal computes fibres. algebraic-lemma-representable-diagonal (cross-chapter, named). Shows the source of the base-changed morphism is an algebraic space.
- Smooth cover criterion. algebraic-lemma-smooth-surjective-morphism-implies-algebraic (cross-chapter, named). Concludes the target fibre product is an algebraic stack.

Source reference. Original reference 0302: focused-seeds/geometry/S9-crit-map-from-algebraic/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 045C: focused-seeds/geometry/S9-crit-map-from-algebraic/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 045G: focused-seeds/geometry/S9-crit-map-from-algebraic/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05UL: focused-seeds/geometry/S9-crit-map-from-algebraic/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $F : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of stacks in groupoids over $(Sch/S)_{fppf}$. If $\mathcal{X}$ is an algebraic stack and $\Delta : \mathcal{Y} \to \mathcal{Y} \times \mathcal{Y}$ is representable by algebraic spaces, then F is algebraic.*

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Proof. Choose a representable stack in groupoids $\mathcal{U}$ and a surjective smooth 1-morphism $\mathcal{U} \to \mathcal{X}$. Let T be a scheme and let ξ be an object of $\mathcal{Y}$ over T. The morphism of 2-fibre products

$$(Sch/T)_{fppf} \times_{\xi,\mathcal{Y}} \mathcal{U} \longrightarrow (Sch/T)_{fppf} \times_{\xi,\mathcal{Y}} \mathcal{X}$$

is representable by algebraic spaces, surjective, and smooth as a base change of $\mathcal{U} \to \mathcal{X}$, see Algebraic Stacks, Lemmas 0302 and 045C. By our condition on the diagonal of $\mathcal{Y}$ we see that the source of this morphism is representable by an algebraic space, see Algebraic Stacks, Lemma 045G. Hence the target is an algebraic stack by Algebraic Stacks, Lemma 05UL. □